

HW1-2 LQR

This notebook completes the three assignment objectives: derive the lateral vehicle model, implement LQR path tracking on multiple paths, and compare controller behavior for different Q/R tuning choices.

Objective 1: Vehicle lateral control state-space model

Use the small-angle linear dynamic bicycle model with constant longitudinal speed v_x . The model uses lateral velocity v_y , yaw rate r , and steering input δ .

The lateral tire slip angles are approximated by

$$\alpha_f \approx \frac{v_y + l_f r}{v_x} - \delta, \quad \alpha_r \approx \frac{v_y - l_r r}{v_x}$$

With linear tire forces,

$$F_{yf} = -C_f \alpha_f = C_f \delta - C_f \frac{v_y + l_f r}{v_x}$$

$$F_{yr} = -C_r \alpha_r = -C_r \frac{v_y - l_r r}{v_x}$$

The lateral force and yaw moment balances are

$$m(\dot{v}_y + v_x r) = F_{yf} + F_{yr}$$

$$I_z \dot{r} = l_f F_{yf} - l_r F_{yr}$$

Substituting the tire forces gives the dynamic bicycle equations:

$$\dot{v}_y = -\frac{C_f + C_r}{mv_x} v_y + \left(\frac{C_r l_r - C_f l_f}{mv_x} - v_x \right) r + \frac{C_f}{m} \delta$$

$$\dot{r} = \frac{C_r l_r - C_f l_f}{I_z v_x} v_y - \frac{C_f l_f^2 + C_r l_r^2}{I_z v_x} r + \frac{C_f l_f}{I_z} \delta$$

For path tracking, augment the dynamic bicycle model with cross-track error and heading error. The signed cross-track error is

$$e = -(x - x_{ref}) \sin \theta_p + (y - y_{ref}) \cos \theta_p$$

and the heading error is defined as

$$\psi_e = \theta - \theta_p$$

With this sign convention and small-angle assumptions,

$$\dot{e} = v_x \psi_e + v_y$$

$$\dot{\psi}_e = r$$

For curved paths, the path-heading rate is handled by curvature feedforward steering; the feedback model above is the local linear tracking model.

Therefore, with state

$$\mathbf{x} = [e, \psi_e, v_y, r]^T$$

and input $u = \delta$, the continuous-time model is

$$\dot{\mathbf{x}} = \mathbf{A}_c \mathbf{x} + \mathbf{B}_c u$$

where

$$\mathbf{A}_c = \begin{bmatrix} 0 & v_x & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -\frac{C_f + C_r}{mv_x} & \frac{C_r l_r - C_f l_f}{mv_x} - v_x \\ 0 & 0 & \frac{C_r l_r - C_f l_f}{I_z v_x} & -\frac{C_f l_f^2 + C_r l_r^2}{I_z v_x} \end{bmatrix}, \quad \mathbf{B}_c = \begin{bmatrix} 0 \\ 0 \\ \frac{C_f}{m} \\ \frac{C_f l_f}{I_z} \end{bmatrix}$$

The model is discretized with forward Euler for the LQR design:

$$\mathbf{A}_d = \mathbf{I} + \mathbf{A}_c \Delta t, \quad \mathbf{B}_d = \mathbf{B}_c \Delta t$$

The discrete LQR control law is

$$\delta = \delta_{ff} - \mathbf{K} \mathbf{x}$$

where $\mathbf{K}$ is found from the discrete algebraic Riccati equation. The curvature feedforward term

$$\delta_{ff} = \tan^{-1}(L\kappa)$$

helps the vehicle follow curved paths without relying only on feedback.

Code

```
%matplotlib inline
```

```

from LQR_Full_state_first_step import (
    continuous_bicycle_matrices,
    lqr_gain,
    run_all_cases,
    print_gain_table,
    print_metric_table,
    make_plots,
    LQRCase,
    simulate_path,
)

A_c, B_c = continuous_bicycle_matrices()
A_c, B_c

```

Output

```

(array([[ 0.          ,  5.          ,  1.          ,  0.          ],
        [ 0.          ,  0.          ,  0.          ,  1.          ],
        [ 0.          ,  0.          , -21.33333333, -0.73333333],
        [ 0.          ,  0.          ,  2.84444444, -28.44444444]]),
array([[ 0.          ],
        [ 0.          ],
        [53.33333333],
        [42.66666667]]))

```

Objective 2: LQR path tracking on different reference paths

The implementation tests the LQR tracker on three reference paths: a polynomial plus sinusoid, a sine path, and a lane-change path. To keep this objective focused on path tracking, the baseline Balanced tuning is used here. The separate Q/R tuning comparison is shown in Objective 3.

For each time step the simulation finds the nearest path point, computes signed cross-track and heading errors, applies the LQR steering command with curvature feedforward and steering saturation, and propagates the nonlinear vehicle position using the dynamic bicycle lateral states.

Code

```

paths, cases, results = run_all_cases()

balanced_case = next(case for case in cases if case.name == "Balanced")

print("Balanced LQR path-tracking metrics")
print("path          rms e    rms psi   max steer   mean |steer rate|")
print("-----")
for path_name in paths:
    result = results[(path_name, balanced_case.name)]
    print(
        f"{{path_name:19s}}"
        f" {{result['rms_e']:.3f}}"
    )

```

```
f" {result['rms_psi_deg']:9.3f}"
f" {result['max_abs_steering_deg']:11.3f}"
f" {result['mean_abs_steering_rate_deg_s']:18.3f}"
)
```

Output

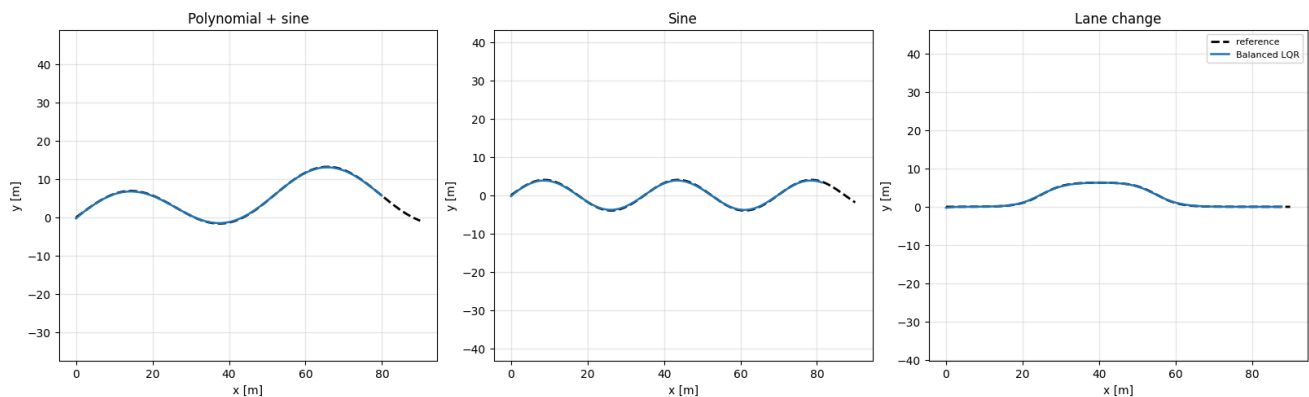
Balanced LQR path-tracking metrics

path	rms e	rms psi	max steer	mean steering rate
Polynomial + sine	0.082	3.684	18.867	27.988
Sine	0.114	5.284	25.082	44.647
Lane change	0.052	2.050	18.137	15.562

Code

```
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 3, figsize=(16, 4.8), constrained_layout=True)
for ax, (path_name, path) in zip(axes, paths.items()):
    ax.plot(path["x"], path["y"], "k--", linewidth=2, label="reference")
    result = results[(path_name, "Balanced")]
    ax.plot(result["x"], result["y"], linewidth=1.8, label="Balanced LQR")
    ax.set_title(path_name)
    ax.set_xlabel("x [m]")
    ax.set_ylabel("y [m]")
    ax.axis("equal")
    ax.grid(True, alpha=0.3)
axes[-1].legend(loc="best", fontsize=8)
plt.show()
```



Interactive LQR simulation

The cell below makes the simulation interactive in Jupyter. The sliders change the main Q penalties and R , while the dropdown selects the reference path. Re-running the cell displays a live plot and metrics for the selected tuning.

Code

```

import numpy as np
import matplotlib.pyplot as plt
from IPython.display import display

from LQR_Full_state_first_step import LQRCase, run_all_cases, simulate_path

try:
    paths
    cases
    results
except NameError:
    paths, cases, results = run_all_cases()

try:
    import ipywidgets as widgets
except ImportError:
    print("Install ipywidgets from requirements.txt to use the interactive simulation.")
else:
    path_dropdown = widgets.Dropdown(options=list(paths.keys()), value="Lane change",
description="Path")
    q_e = widgets.FloatLogSlider(value=30.0, base=10, min=0, max=3, step=0.05,
description="Q e")
    q_psi = widgets.FloatLogSlider(value=45.0, base=10, min=0, max=3, step=0.05,
description="Q psi")
    r_weight = widgets.FloatLogSlider(value=1.0, base=10, min=-1, max=1.2, step=0.05,
description="R")

    def run_interactive(path_name, q_e, q_psi, r_weight):
        case = LQRCase("Interactive", (q_e, q_psi, 1.0, 1.0), r_weight)
        result = simulate_path(paths[path_name], case)
        fig, axes = plt.subplots(1, 2, figsize=(12, 4), constrained_layout=True)
        axes[0].plot(paths[path_name]["x"], paths[path_name]["y"], "k--",
label="reference")
        axes[0].plot(result["x"], result["y"], label="vehicle")
        axes[0].axis("equal")
        axes[0].grid(True, alpha=0.3)
        axes[0].legend()
        axes[0].set_title(path_name)
        axes[1].plot(result["t"], result["e"], label="cross-track error [m]")
        axes[1].plot(result["t"], np.rad2deg(result["steer"]), label="steering [deg]")
        axes[1].grid(True, alpha=0.3)
        axes[1].legend()
        plt.show()
        print(f"RMS e: {result['rms_e']:.3f} m")
        print(f"RMS heading error: {result['rms_psi_deg']:.3f} deg")
        print(f"Max steering: {result['max_abs_steer_deg']:.3f} deg")
        print(f"Mean |steer rate|: {result['mean_abs_steer_rate_deg_s']:.3f} deg/s")

    controls = {
        "path_name": path_dropdown,
        "q_e": q_e,
        "q_psi": q_psi,
        "r_weight": r_weight,
    }
    display(widgets.HBox([path_dropdown, q_e, q_psi, r_weight]))
    display(widgets.interactive_output(run_interactive, controls))

```

Output

```
HBox(children=(Dropdown(description='Path', index=2, options=('Polynomial + sine',
'Sine', 'Lane change'), val...
```

Output

```
Output()
```

Objective 3: Q/R tuning observations

The tuning cases below change the diagonal Q penalties and scalar R penalty to show the expected LQR tradeoffs. The gain table shows how each tuning changes the feedback weights, and the metric table compares tracking error, heading error, steering magnitude, and steering smoothness on every path.

- **Balanced** uses moderate state penalties and gives a reasonable compromise between tracking and steering effort. It does not produce the smallest error, but it avoids the highest steering demands.
- **Aggressive** increases the e and ψ_e penalties while reducing R, so tracking improves compared with Balanced. The cost is larger and faster steering: for example, on the sine path the mean absolute steering rate increases from about 44.6 deg/s for Balanced to about 64.3 deg/s for Aggressive.
- **Smooth steering** increases R, so steering effort is penalized more heavily. This reduces steering activity but allows larger tracking error. It has the lowest steering-rate value on each path, such as 12.6 deg/s on the lane-change path, but its RMS cross-track error is consistently higher than Balanced.
- **Lateral-error focused** heavily penalizes cross-track error. This produced the best RMS cross-track tracking on all three paths: 0.030 m for lane change, 0.040 m for polynomial plus sine, and 0.055 m for sine. The tradeoff is more aggressive steering, including saturation at 35 deg on the lane-change path.
- **Heading-error focused** heavily penalizes heading error. This gave the lowest RMS heading error on all three paths, but it also produced the largest RMS cross-track error in each case. This shows that aligning the vehicle heading with the path does not automatically minimize lateral displacement from the path.

Overall, increasing entries in Q makes the corresponding state error more expensive, while increasing R makes steering more expensive. The best overall choice depends on the design goal: use the lateral-error focused tuning for minimum path error, the heading-

error focused tuning for best yaw alignment, the smooth-steering tuning for actuator/passenger comfort, and the balanced tuning when no single metric should dominate.

Code

```
from LQR_Full_state_first_step import run_all_cases, print_gain_table,
print_metric_table

try:
    paths
    cases
    results
except NameError:
    paths, cases, results = run_all_cases()

print_gain_table(cases)
print_metric_table(results, cases)
```

Output

LQR gains for $x = [e, \psi_e, v_y, r]^T$

case	k_e	k_ψ	k_{vy}	k_r
Balanced	1.265	2.078	0.077	0.001
Aggressive	2.279	2.761	0.125	0.024
Smooth steering	0.876	1.707	0.054	0.009
Lateral-error focused	2.712	2.415	0.141	0.014
Heading-error focused	1.095	2.886	0.074	0.032

Tracking metrics

path rate	case	rms e	rms ψ	max steer	mean steer
-----	-----	-----	-----	-----	-----
Lane change	Balanced	0.052	2.050	18.137	
15.562					
Lane change	Aggressive	0.037	2.096	32.659	
23.223					
Lane change	Smooth steering	0.062	2.070	12.564	
12.610					
Lane change	Lateral-error focused	0.030	2.168	35.000	
21.475					
Lane change	Heading-error focused	0.075	1.915	15.695	
20.598					
Polynomial + sine	Balanced	0.082	3.684	18.867	
27.988					
Polynomial + sine	Aggressive	0.056	3.702	26.332	
40.965					
Polynomial + sine	Smooth steering	0.098	3.696	15.383	
22.903					
Polynomial + sine	Lateral-error focused	0.040	3.789	31.320	
36.430					
Polynomial + sine	Heading-error focused	0.127	3.501	16.437	
39.484					
Sine	Balanced	0.114	5.284	25.082	
44.647					

Sine 64.319	Aggressive	0.080	5.421	26.003
Sine 36.869	Smooth steering	0.137	5.329	24.019
Sine 58.120	Lateral-error focused	0.055	5.475	31.044
Sine 62.852	Heading-error focused	0.172	4.884	25.270

Code

```

import numpy as np
import matplotlib.pyplot as plt
from LQR_Full_state_first_step import run_all_cases

try:
    paths
    cases
    results
except NameError:
    paths, cases, results = run_all_cases()

fig, axes = plt.subplots(len(paths), 2, figsize=(13, 10), sharex=False,
constrained_layout=True)
for row, (path_name, _) in enumerate(paths.items()):
    for case in cases:
        result = results[(path_name, case.name)]
        axes[row, 0].plot(result["t"], result["e"], label=case.name)
        axes[row, 1].plot(result["t"], np.rad2deg(result["steer"]), label=case.name)
    axes[row, 0].set_title(f"{path_name}: cross-track error")
    axes[row, 1].set_title(f"{path_name}: steering")
    axes[row, 0].set_ylabel("e [m]")
    axes[row, 1].set_ylabel("delta [deg]")
    axes[row, 0].grid(True, alpha=0.3)
    axes[row, 1].grid(True, alpha=0.3)
axes[-1, 0].set_xlabel("time [s]")
axes[-1, 1].set_xlabel("time [s]")
axes[0, 1].legend(loc="best", fontsize=8)
plt.show()

```

